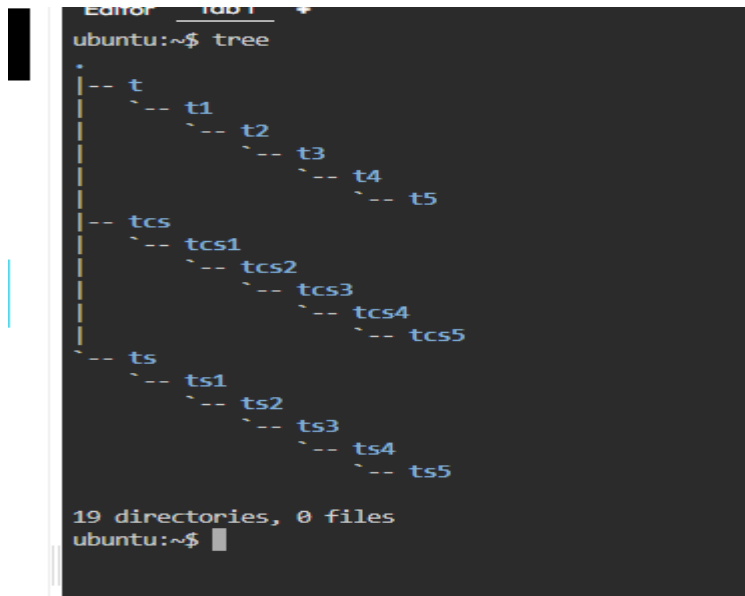


Linux Commands (S3)

- **mv** - change filename or directoriesname

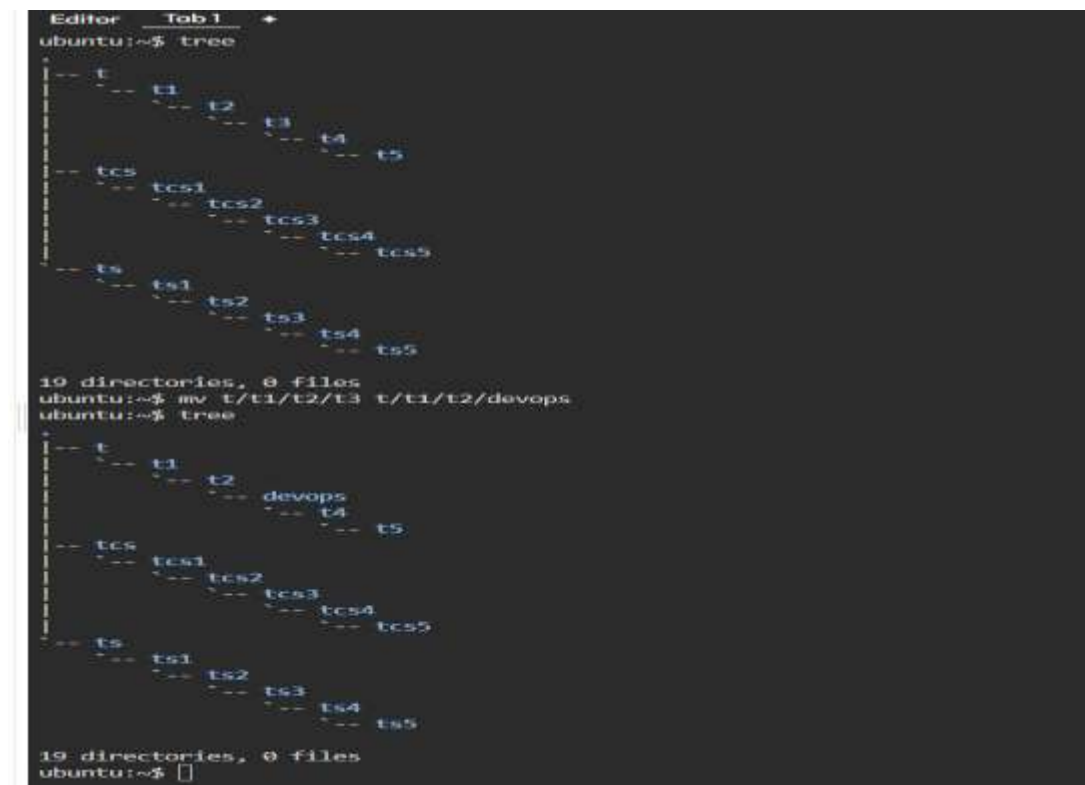
Syntax : `mv sourcefile destinationfile`

Example 1:

A terminal window titled 'Editor Tab 1' showing the output of the 'tree' command in a directory named 't'. The directory structure is as follows: 't' contains 't1', 't2', 't3', 't4', and 't5'. 't1' contains 'tcs1', 'tcs2', 'tcs3', 'tcs4', and 'tcs5'. 't2' contains 'ts1', 'ts2', 'ts3', 'ts4', and 'ts5'. The terminal shows '19 directories, 0 files' and the prompt 'ubuntu:~\$'.

I want change t3 name into devops

Syntax : `mv t/t1/t2/t3 t/t1/t2/devops`

A terminal window titled 'Editor Tab 1' showing the output of the 'tree' command after the 'mv' command. The directory structure is now: 't' contains 't1', 't2', 'devops', 't4', and 't5'. 't1' contains 'tcs1', 'tcs2', 'tcs3', 'tcs4', and 'tcs5'. 't2' contains 'ts1', 'ts2', 'ts3', 'ts4', and 'ts5'. The terminal shows '19 directories, 0 files' and the prompt 'ubuntu:~\$'.

Example 2:

I want move tcs3 into ts3

Syntax : `mv tcs/tcs1/tcs2/tcs3 ts/ts1/ts3`

```
Editor  Tab 1
ubuntu:~$ tree
.
|-- t
|   |-- t1
|   |   |-- t2
|   |   |   |-- devops
|   |   |   |   |-- t4
|   |   |   |   |-- t5
|   |-- tcs
|   |   |-- tcs1
|   |   |   |-- tcs2
|   |   |   |   |-- tcs3
|   |   |   |   |-- tcs4
|   |   |   |   |-- tcs5
|   |-- ts
|   |   |-- ts1
|   |   |   |-- ts2
|   |   |   |   |-- ts3
|   |   |   |   |-- ts4
|   |   |   |   |-- ts5
|   |-- tcs4
|   |-- tcs5
|-- ts
|   |-- ts1
|   |   |-- ts2
|   |   |   |-- ts3
|   |   |   |-- ts4
|   |   |   |-- ts5
|   |-- tcs4
|   |-- tcs5
|-- ts4
|-- ts5
|-- tcs4
|-- tcs5
19 directories, 0 files
ubuntu:~$ mv tcs/tcs1/tcs2/tcs3 ts/ts1/ts3
ubuntu:~$ tree
.
|-- t
|   |-- t1
|   |   |-- t2
|   |   |   |-- devops
|   |   |   |   |-- t4
|   |   |   |   |-- t5
|   |-- tcs
|   |   |-- tcs1
|   |   |   |-- tcs2
|   |-- ts
|   |   |-- ts1
|   |   |   |-- ts2
|   |   |   |   |-- ts3
|   |   |   |   |-- ts4
|   |   |   |   |-- ts5
|   |-- tcs4
|   |-- tcs5
|-- ts
|   |-- ts1
|   |   |-- ts2
|   |   |   |-- ts3
|   |   |   |-- ts4
|   |   |   |-- ts5
|   |-- tcs4
|   |-- tcs5
19 directories, 0 files
ubuntu:~$
```

- **cp** - copy files or directories

Syntax : `cp -r sourcefile destinationfile`

Example 1:

I want copy devops into tcs2

Syntax : `cp -r t/t1/t2/devops tcs/tcs1/tcs2`

```

-- t
  -- t1
    -- t2
      -- devops
      -- t4
      -- t5
  -- tcs
    -- tcs1
      -- tcs2
  -- ts
    -- ts1
      -- ts2
      -- ts3
      -- ts4
      -- ts5
    -- ts3
      -- tcs4
      -- tcs5

19 directories, 0 files
ubuntu:~$ cp -r t/t1/t2/devops tcs/tcs1/tcs2
ubuntu:~$ tree
-- t
  -- t1
    -- t2
      -- devops
      -- t4
      -- t5
  -- tcs
    -- tcs1
      -- tcs2
        -- devops
        -- t4
        -- t5
  -- ts
    -- ts1
      -- ts2
      -- ts3
      -- ts4
      -- ts5
    -- ts3
      -- tcs4
      -- tcs5

22 directories, 0 files
ubuntu:~$

```

Git

Git : Global Information Tracker

Git three-stage architecture



The Files that are present in the Working Directory git will not track

Check if Git is Install or not

Syntax : `git --version`

```
Editor  Tab 1  +
ubuntu:~$ git --version
git version 2.43.0
ubuntu:~$
```

Now I want to create files

Syntax : `touch tcs pwc ghc`

```
Editor  Tab1  +
ubuntu:~$ touch tcs pwc ghc
ubuntu:~$ ls
filesystem ghc pwc tcs
ubuntu:~$
```

If you need to track files, the files are saved into the `.git` folder.

When installing Git, the `.git` folder is created.

Syntax : `git init`

```
ubuntu:~$ ll
total 36
drwx----- 4 root root 4096 Oct 16 18:49 ./
drwxr-xr-x 22 root root 4096 Sep 19 18:23 ../
-rw----- 1 root root 10 Feb 10 2025 .bash_history
-rw-r--r-- 1 root root 3194 Sep 19 18:23 .bashrc
-rw-r--r-- 1 root root 161 Apr 22 2024 .profile
drwx----- 2 root root 4096 Sep 19 18:21 .ssh/
drwxr-xr-x 4 root root 4096 Oct 16 18:45 .theia/
-rw-r--r-- 1 root root 109 Sep 19 18:23 .vimrc
-rw-r--r-- 1 root root 165 Sep 19 18:23 .wget-hsts
lrwxrwxrwx 1 root root 1 Sep 19 18:23 filesystem -> //
-rw-r--r-- 1 root root 0 Oct 16 18:49 ghc
-rw-r--r-- 1 root root 0 Oct 16 18:49 pwc
-rw-r--r-- 1 root root 0 Oct 16 18:49 tcs
ubuntu:~$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /root/.git/
ubuntu:~$ ll
total 40
drwx----- 5 root root 4096 Oct 16 18:53 ./
drwxr-xr-x 22 root root 4096 Sep 19 18:23 ../
-rw----- 1 root root 10 Feb 10 2025 .bash_history
-rw-r--r-- 1 root root 3194 Sep 19 18:23 .bashrc
drwxr-xr-x 7 root root 4096 Oct 16 18:53 .git/
-rw-r--r-- 1 root root 161 Apr 22 2024 .profile
drwx----- 2 root root 4096 Sep 19 18:21 .ssh/
drwxr-xr-x 4 root root 4096 Oct 16 18:45 .theia/
-rw-r--r-- 1 root root 109 Sep 19 18:23 .vimrc
-rw-r--r-- 1 root root 165 Sep 19 18:23 .wget-hsts
lrwxrwxrwx 1 root root 1 Sep 19 18:23 filesystem -> //
-rw-r--r-- 1 root root 0 Oct 16 18:49 ghc
-rw-r--r-- 1 root root 0 Oct 16 18:49 pwc
-rw-r--r-- 1 root root 0 Oct 16 18:49 tcs
ubuntu:~$
```

```
Editor  Tab1  +
ubuntu:~$ ls
filesystem ghc pwc tcs
ubuntu:~$
```

When I have three files, I can move them from the working area to the staging area.

Syntax : `git add filename`

When I want to check whether my file is in the working area or the staging area.

Syntax : `git status`

```
Editor  Tab1  +
ubuntu:~$ ls
filesystem ghc pwc tcs
ubuntu:~$ git add pwc
ubuntu:~$ git status
On branch master

No commits yet

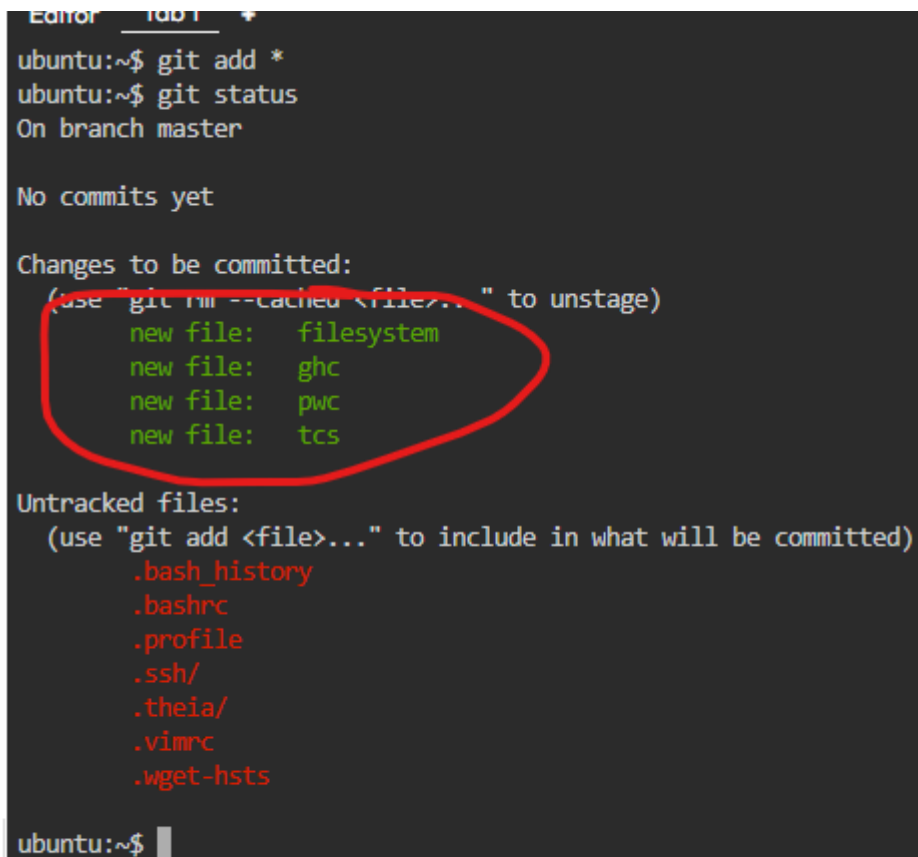
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   pwc

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .bash_history
    .bashrc
    .profile
    .ssh/
    .theia/
    .vimrc
    .wget-hsts
    filesystem
    ghc
    tcs

ubuntu:~$
```

When I want to move all files from the working area to the staging area.

Syntax : `git add *`



```
Editor  Tab 1
ubuntu:~$ git add *
ubuntu:~$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   filesystem
    new file:   ghc
    new file:   pwc
    new file:   tcs

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .bash_history
    .bashrc
    .profile
    .ssh/
    .theia/
    .vimrc
    .wget-hsts

ubuntu:~$
```

When I want to remove tcs files from staging area.

Syntax : `git rm --cached filename(tcs)`

```
ubuntu:~$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   filesystem
    new file:   ghc
    new file:   pwc
    new file:   tcs

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .bash_history
    .bashrc
    .profile
    .ssh/
    .theia/
    .vimrc
    .wget-hsts

ubuntu:~$ git rm --cached tcs
rm 'tcs'
ubuntu:~$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   filesystem
    new file:   ghc
    new file:   pwc

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .bash_history
    .bashrc
    .profile
    .ssh/
    .theia/
    .vimrc
    .wget-hsts
    tcs

ubuntu:~$
```


configure your username and email:

Syntax :

git config --global user.name "Your Name"

git config --global user.email "your@email.com"

How to check configure or not

Syntax :

git config --list

```
Editor  Tab 1  +
ubuntu:~$ git config --global user.name "Rajesh"
ubuntu:~$ git config --global user.email "rajesh45@gmail.com"
ubuntu:~$ git config --list
user.name=Rajesh
user.email=rajesh45@gmail.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
ubuntu:~$
```

I want to move files from staging area to repository(it means save files)

Syntax : **git commit -m "message" file name**

```
Editor  Tab 1  +
ubuntu:~$ git commit -m "pwc 1st data" pwc
[master (root-commit) b682a78] pwc 1st data
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 pwc
ubuntu:~$
```

How to check the file commit or not

Syntax : `git log`

```
Editor  Tab1  +
ubuntu:~$ git commit -m "pwc 1st data" pwc
[master (root-commit) b682a78] pwc 1st data
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 pwc
ubuntu:~$ git log
commit b682a78e95cad9b306d60833b0278d97c045da38 (HEAD -> master)
Author: Rajesh <rajesh45@gmail.com>
Date: Thu Oct 16 19:20:21 2025 +0000

    pwc 1st data
ubuntu:~$
```

I need with File name

Syntax : `git log --name-only`

```
Editor  Tab1  +
ubuntu:~$ git commit -m "pwc 1st data" pwc
[master (root-commit) b682a78] pwc 1st data
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 pwc
ubuntu:~$ git log
commit b682a78e95cad9b306d60833b0278d97c045da38 (HEAD -> master)
Author: Rajesh <rajesh45@gmail.com>
Date: Thu Oct 16 19:20:21 2025 +0000

    pwc 1st data
ubuntu:~$ git log --name-only
commit b682a78e95cad9b306d60833b0278d97c045da38 (HEAD -> master)
Author: Rajesh <rajesh45@gmail.com>
Date: Thu Oct 16 19:20:21 2025 +0000

    pwc 1st data
    pwc
ubuntu:~$
```